

3.ORTHO POWER DRILL

TECHNICAL SPECIFICATION

BATTERY-OPERATED ORTHOPAEDIC POWER TOOL SYSTEM

Intended Application: Orthopaedic Trauma, Joint Reconstruction, Total Hip Replacement (THR), Total Knee Replacement (TKR), Intramedullary Nailing, Bone Reaming, Drilling and Sawing.

A. Main System Specifications

Sl. No.	Parameter	Minimum Tender Requirement
1	System Type	High-performance battery-operated orthopaedic power tool system
2	Applications	Drilling, reaming, sawing, K-wire/pin insertion and removal, and bone cutting
3	System Configuration	Modular system with dedicated battery-powered handpieces
4	Handpieces	Minimum 3 dedicated handpieces for drilling/reaming, oscillating sawing and reciprocating sawing
5	Battery Type	Rechargeable Lithium-ion battery
6	Battery Voltage	Approximately 14.8 V or equivalent
7	Battery Capacity	Approximately 1.5 Ah / 22.2 Wh or better
8	Charging Time	Typically less than 60 minutes
9	Sterility	Battery system shall be suitable for validated sterile surgical use with appropriate sterile battery casing/cover
10	Battery Compatibility	Battery and handpieces shall be designed for reliable operation and secure coupling
11	Ergonomics	Ergonomically designed handpieces suitable for prolonged orthopaedic surgical procedures
12	Activation	Variable speed control through handpiece trigger
13	Direction Control	Forward and reverse rotation shall be available where applicable
14	Safety Lock	Safety locking mechanism to prevent unintended activation
15	Cannulation	Reamer/Drill handpiece shall provide cannulation of approximately 4 mm or better
16	Coupling	Quick and secure coupling system for interchangeable attachments
17	Attachments	Wide range of dedicated attachments for drilling, reaming, sawing and fixation procedures

Sl. No.	Parameter	Minimum Tender Requirement
18	Sterilization	Handpieces and reusable attachments shall be compatible with validated hospital sterilization procedures according to manufacturer's IFU
19	Cleaning	Components shall be designed for effective cleaning and reprocessing
20	Safety	Appropriate electrical, mechanical and operational safety features shall be incorporated

B. Battery Reamer/Drill Handpiece

Parameter	Minimum Requirement
Speed	0–340 RPM , without attachment; maximum speed may vary depending on attachment
Torque	0–15 Nm , with maximum torque dependent on attachment
Cannulation	4.0 mm diameter or better
Weight	Approximately 1,565 g including battery , or lower
Application	High-torque drilling and reaming for large-bone orthopaedic procedures
Control	Variable speed trigger
Rotation	Forward and reverse
Attachments	Compatible with reaming, drilling, AO/ASIF and other required orthopaedic attachments

C. Battery Oscillator Handpiece

Parameter	Minimum Requirement
Speed	Up to 12,000 oscillations/minute
Deflection	Approximately 4.5 mm
Weight	Approximately 1,685 g including battery , or lower
Application	Oscillating bone sawing and cutting
Blade Compatibility	Compatible with a comprehensive range of large-bone saw blades
Control	Variable speed trigger
Safety	Controlled blade activation and secure blade locking mechanism

D. Battery Reciprocator Handpiece

Parameter	Minimum Requirement
Speed	Up to 14,000 oscillations/minute
Stroke	Approximately 4 mm
Weight	Approximately 1,675 g including battery , or lower

Parameter	Minimum Requirement
Application	Reciprocating bone cutting and sawing
Blade Compatibility	Compatible with suitable reciprocating saw blades
Control	Variable speed trigger
Safety	Secure blade locking and controlled activation

E. Required Attachments and Accessories

The complete system shall include, as a minimum:

1. Battery Reamer/Drill Handpiece – **1 No.**
2. Battery Oscillator Handpiece – **1 No.**
3. Battery Reciprocator Handpiece – **1 No.**
4. Rechargeable Li-ion Battery – minimum **2 Nos.**
5. Sterile Battery Casing – suitable quantity
6. Sterile Battery Covers – suitable quantity
7. Battery Reamer/Drill Attachments – suitable range
8. AO/ASIF Quick Coupling Attachment – **1 No.**
9. Large Bone Reaming Attachments – suitable range
10. Drill Attachments – suitable range
11. Oscillating Saw Attachments – suitable range
12. Reciprocating Saw Attachments – suitable range
13. K-wire/Pin Driver Attachment – **1 No.**
14. Pin Driver suitable for **3.2 mm pins** – **1 No.**, if required
15. Saw Blades – suitable initial assortment
16. Drill Bits and Reamers – suitable initial assortment
17. Sterilization Tray/Container – **1 complete set**
18. Universal Battery Charger – **1 No.**
19. Required power cable and accessories
20. Manufacturer's recommended maintenance/lubrication accessories
21. User Manual – **1 set**
22. Service Manual – **1 set**

F. Battery Charger

The charger shall:

- Li-ion batteries.
- Provide safe and controlled battery charging.
- Have independent charging bays.
- Provide charging status indication.
- Support simultaneous charging of multiple batteries.
- Include necessary power cord and accessories.

G. Tender Requirements

Parameter	Requirement
Installation	Installation and commissioning at hospital site
Demonstration	Live demonstration to technical evaluation committee, if requested
Training	Training for Orthopaedic Surgeons, OT Nurses and Biomedical Engineering Team
Warranty	Minimum 3 years comprehensive warranty from installation/commissioning
Service	Authorized service support in India
Spare Parts	Availability of genuine spare parts for minimum 7–10 years
Preventive Maintenance	As per manufacturer's recommended schedule
Documentation	User manual, service manual, IFU and sterilization guidelines
Regulatory	Compliance with applicable Indian regulatory requirements
Safety	Compliance with applicable medical electrical safety standards
Technical Bid	Manufacturer's original technical datasheet to be submitted
Compliance Statement	Point-by-point compliance statement mandatory

Warranty & After-Sales Support

- **Comprehensive Warranty:** Minimum 3 years from the date of installation/commissioning.
- **Preventive Maintenance:** Minimum 4 preventive maintenance visits per year during the warranty period.
- **Spare Parts Support:** Availability and support for genuine spare parts for a minimum period of 10 years.
- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.